

Assignment-01: Computer Networks – Windows Commands

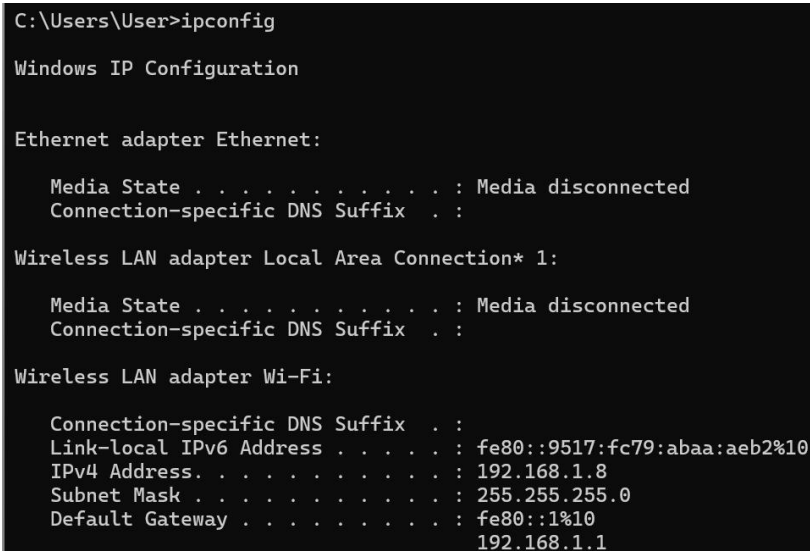
Name: Abdullah Ayub
Roll NO: BSDSF24M029

IP & Network Configuration

Command: ipconfig

Description: Displays basic TCP/IP configuration information including IPv4 and IPv6 addresses of network adapters.

Screenshot:



```
C:\Users\User>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9517:fc79:abaa:aeb2%10
    IPv4 Address. . . . . : 192.168.1.8
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%10
                                192.168.1.1
```

Command: ipconfig /all

Description: Displays detailed network configuration such as MAC address, DNS servers, DHCP status, and lease information.

Screenshot:

```
C:\Users\User>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : Mian-Abdullah
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (23) I219-V
Physical Address. . . . . : 84-BA-59-8B-FC-58
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 72-08-94-C6-78-7B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 70-08-94-C6-78-7B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::9517:fc79:abaa:aeb2%10(Preferred)
IPv4 Address. . . . . : 192.168.1.8(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, February 5, 2026 3:02:21 PM
Lease Expires . . . . . : Friday, February 6, 2026 3:02:22 PM
Default Gateway . . . . . : fe80::1%10
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 108005524
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-12-DF-33-84-BA-59-8B-FC-58
DNS Servers . . . . . : 192.168.1.1
Primary WINS Server . . . . . : 192.168.1.1
Secondary WINS Server . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

Command: ipconfig /release

Description: Releases the current IP address assigned by the DHCP server.

Screenshot:

```
C:\Users\User>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9517:fc79:abaa:aeb2%10
    Default Gateway . . . . . : fe80::1%10
```

Command: ipconfig /renew

Description: Requests a new IP address from the DHCP server.

Screenshot:

```
C:\Users\User>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9517:fc79:abaa:aeb2%10
    IPv4 Address. . . . . : 192.168.1.8
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%10
                                192.168.1.1
```

Command: ipconfig /flushdns

Description: Clears the DNS resolver cache to remove outdated DNS records.

Screenshot:

```
C:\Users\User>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
```

Command: ipconfig /displaydns

Description: Displays cached DNS entries stored on the local computer.

Screenshot:

```
C:\Users\User>ipconfig /displaydns

Windows IP Configuration

    ww9.fitgirl-repacks.xyz
    -----
    No records of type AAAA

    Record Name . . . . . : ww9.fitgirl-repacks.xyz
    Record Type . . . . . : 1
    Time To Live . . . . . : 604800
    Data Length . . . . . : 4
    Section . . . . . : Answer
    A (Host) Record . . . : 109.94.209.70

    *.fitgirl-repacks.xyz
    -----
    No records of type AAAA

    Record Name . . . . . : *.fitgirl-repacks.xyz
    Record Type . . . . . : 1
    Time To Live . . . . . : 604800
    Data Length . . . . . : 4
    Section . . . . . : Answer
```

Connectivity Testing

Command: ping www.google.com

Description: Tests connectivity between the local system and a remote host.

Screenshot:

```
C:\Users\User>ping www.google.com

Pinging www.google.com [74.125.90.68] with 32 bytes of data:
Reply from 74.125.90.68: bytes=32 time=88ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115
Reply from 74.125.90.68: bytes=32 time=35ms TTL=115

Ping statistics for 74.125.90.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 33ms, Maximum = 88ms, Average = 47ms
```

Command: ping -t www.google.com

Description: Sends continuous ICMP echo requests until manually stopped.

Screenshot:

```
C:\Users\User>ping -t www.google.com

Pinging www.google.com [74.125.90.68] with 32 bytes of data:
Reply from 74.125.90.68: bytes=32 time=92ms TTL=115
Reply from 74.125.90.68: bytes=32 time=39ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=32ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=39ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=32ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=41ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115
Reply from 74.125.90.68: bytes=32 time=32ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115
```

```
Reply from 74.125.90.68: bytes=32 time=33ms TTL=115
Reply from 74.125.90.68: bytes=32 time=32ms TTL=115
Reply from 74.125.90.68: bytes=32 time=35ms TTL=115
Reply from 74.125.90.68: bytes=32 time=35ms TTL=115
Reply from 74.125.90.68: bytes=32 time=37ms TTL=115
Reply from 74.125.90.68: bytes=32 time=34ms TTL=115

Ping statistics for 74.125.90.68:
    Packets: Sent = 111, Received = 111, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 31ms, Maximum = 92ms, Average = 35ms
Control-C
^C
```

Stopped using ctrl+c.

Command: ping -l 1024 www.google.com

Description: Sends ICMP packets with a specified packet size.

Screenshot:

```
C:\Users\User>ping -l 1024 www.google.com

Pinging www.google.com [74.125.90.68] with 1024 bytes of data:
Reply from 74.125.90.68: bytes=1024 time=176ms TTL=115
Reply from 74.125.90.68: bytes=1024 time=35ms TTL=115
Reply from 74.125.90.68: bytes=1024 time=70ms TTL=115
Reply from 74.125.90.68: bytes=1024 time=35ms TTL=115

Ping statistics for 74.125.90.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 35ms, Maximum = 176ms, Average = 79ms
```

Command: ping -a 192.168.1.1

Description: Resolves an IP address to a hostname while testing connectivity.

Screenshot:

```
C:\Users\User>ping -a 192.168.1.1

Pinging MYGROUP [192.168.1.1] with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=64ms TTL=64
Reply from 192.168.1.1: bytes=32 time=6ms TTL=64
Reply from 192.168.1.1: bytes=32 time=7ms TTL=64
Reply from 192.168.1.1: bytes=32 time=7ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 64ms, Average = 21ms
```

Command: tracert www.google.com

Description: Displays the route packets take to reach the destination.

Screenshot:

```
C:\Users\User>tracert www.google.com

Tracing route to www.google.com [74.125.90.68]
over a maximum of 30 hops:

  0  5 ms    2 ms    7 ms    MYGROUP [192.168.1.1]
  1  *        5 ms    *        10.0.0.1
  2  97 ms    *        5 ms    51.198.zcomnetworks.com.pk [103.115.198.51]
  3  6 ms     *        5 ms    110.93.205.184
  4  22 ms    7 ms    12 ms    110.93.254.33
  5  50 ms    26 ms    22 ms    110.93.254.38
  6  104 ms   21 ms    22 ms    110.93.252.136
  7  *        33 ms    36 ms    110.93.192.111
  8  100 ms   34 ms    35 ms    216.239.41.83
  9  96 ms    36 ms    67 ms    192.178.250.97
 10  34 ms    34 ms    37 ms    lcmcta-al-in-f4.1e100.net [74.125.90.68]

Trace complete.
```

Command: pathping www.google.com

Description: Analyzes packet loss and latency along a network path.

Screenshot:

```
C:\Users\User>pathping www.google.com

Tracing route to www.google.com [74.125.90.68]
over a maximum of 30 hops:
 0  Mian-Abdullah [192.168.1.8]
 1  MYGROUP [192.168.1.1]
 2  * * 10.0.0.1
 3  51.198.zcomnetworks.com.pk [103.115.198.51]
 4  110.93.205.184
 5  110.93.254.33
 6  110.93.254.38
 7  110.93.252.136
 8  110.93.192.111
 9  216.239.41.83
10  192.178.250.97
11  lcmcta-al-in-f4.1e100.net [74.125.90.68]

Computing statistics for 275 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				Mian-Abdullah [192.168.1.8]
1	6ms	1/ 100 = 1%	1/ 100 = 1%	MYGROUP [192.168.1.1]
2	13ms	0/ 100 = 0%	0/ 100 = 0%	10.0.0.1
3	13ms	0/ 100 = 0%	0/ 100 = 0%	51.198.zcomnetworks.com.pk [103.115.198.51]
4	9ms	0/ 100 = 0%	0/ 100 = 0%	110.93.205.184
5	12ms	0/ 100 = 0%	0/ 100 = 0%	110.93.254.33
6	25ms	2/ 100 = 2%	2/ 100 = 2%	110.93.254.38
7	29ms	0/ 100 = 0%	0/ 100 = 0%	110.93.252.136
8	36ms	0/ 100 = 0%	0/ 100 = 0%	110.93.192.111
9	---	100/ 100 =100%	100/ 100 =100%	216.239.41.83
10	39ms	0/ 100 = 0%	0/ 100 = 0%	192.178.250.97
11	34ms	1/ 100 = 1%	0/ 100 = 0%	lcmcta-al-in-f4.1e100.net [74.125.90.68]

Network Statistics & Connections

Command: netstat

Description: Displays active TCP network connections.

Screenshot:

```
C:\Users\User>netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49688	Mian-Abdullah:49689	ESTABLISHED
TCP	127.0.0.1:49689	Mian-Abdullah:49688	ESTABLISHED
TCP	127.0.0.1:49690	Mian-Abdullah:49691	ESTABLISHED
TCP	127.0.0.1:49691	Mian-Abdullah:49690	ESTABLISHED
TCP	127.0.0.1:49692	Mian-Abdullah:49693	ESTABLISHED
TCP	127.0.0.1:49693	Mian-Abdullah:49692	ESTABLISHED
TCP	127.0.0.1:51752	Mian-Abdullah:51753	ESTABLISHED
TCP	127.0.0.1:51753	Mian-Abdullah:51752	ESTABLISHED
TCP	192.168.1.8:49896	whatsapp-cdn-shv-01-mct1:5222	ESTABLISHED
TCP	192.168.1.8:49897	172.64.148.235:https	ESTABLISHED
TCP	192.168.1.8:50785	246:https	CLOSE_WAIT
TCP	192.168.1.8:50786	119.63.197.151:http	FIN_WAIT_1
TCP	192.168.1.8:50787	119.63.197.151:http	FIN_WAIT_1
TCP	192.168.1.8:50788	113.44.253.43:http	TIME_WAIT
TCP	192.168.1.8:50789	119.63.197.151:http	FIN_WAIT_1
TCP	192.168.1.8:50790	pnfjra-ap-in-f14:https	ESTABLISHED
TCP	192.168.1.8:50791	119.63.197.151:http	FIN_WAIT_1
TCP	192.168.1.8:54292	246:https	CLOSE_WAIT
TCP	192.168.1.8:54349	pnfjra-ap-in-f14:https	TIME_WAIT
TCP	192.168.1.8:54350	pnfjra-ap-in-f14:https	TIME_WAIT
TCP	192.168.1.8:54351	pnfjra-ap-in-f14:https	TIME_WAIT
TCP	192.168.1.8:54355	113.44.253.43:http	TIME_WAIT
TCP	192.168.1.8:54358	119.63.197.151:http	FIN_WAIT_1
TCP	192.168.1.8:54359	119.63.197.151:http	FIN_WAIT_1
TCP	192.168.1.8:54360	119.63.197.151:http	FIN_WAIT_1

Command: netstat -an

Description: Displays all network connections using numerical addresses and port numbers.

Screenshot:

```
C:\Users\User>netstat -an
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49676	0.0.0.0:0	LISTENING
TCP	127.0.0.1:2680	0.0.0.0:0	LISTENING
TCP	127.0.0.1:27015	0.0.0.0:0	LISTENING
TCP	127.0.0.1:27182	0.0.0.0:0	LISTENING
TCP	127.0.0.1:49688	127.0.0.1:49689	ESTABLISHED
TCP	127.0.0.1:49689	127.0.0.1:49688	ESTABLISHED
TCP	127.0.0.1:49690	127.0.0.1:49691	ESTABLISHED
TCP	127.0.0.1:49691	127.0.0.1:49690	ESTABLISHED
TCP	127.0.0.1:49692	127.0.0.1:49693	ESTABLISHED
TCP	127.0.0.1:49693	127.0.0.1:49692	ESTABLISHED
TCP	127.0.0.1:51752	127.0.0.1:51753	ESTABLISHED
TCP	127.0.0.1:51753	127.0.0.1:51752	ESTABLISHED
TCP	127.0.0.1:56842	0.0.0.0:0	LISTENING
TCP	127.0.0.1:63289	0.0.0.0:0	LISTENING
TCP	127.0.0.1:63295	0.0.0.0:0	LISTENING


```

UDP    0.0.0.0:5353          *: *
UDP    0.0.0.0:5353          *: *
UDP    0.0.0.0:5355          *: *
UDP    0.0.0.0:53036        172.64.146.215:443
UDP    0.0.0.0:53392        142.250.186.234:443
UDP    0.0.0.0:53920        142.250.186.234:443
UDP    0.0.0.0:55350        142.250.186.234:443
UDP    127.0.0.1:1900        *: *
UDP    127.0.0.1:49664      127.0.0.1:49665
UDP    127.0.0.1:49665      127.0.0.1:49664
UDP    127.0.0.1:49666      127.0.0.1:49666
UDP    127.0.0.1:54242      *: *
UDP    192.168.1.8:137      *: *
UDP    192.168.1.8:138      *: *
UDP    192.168.1.8:1900     *: *
UDP    192.168.1.8:2177     *: *
UDP    192.168.1.8:54241    *: *
UDP    [::]:123              *: *
UDP    [::]:5353             *: *
UDP    [::]:5353             *: *
UDP    [::]:5353             *: *
UDP    [::]:5353             *: *
UDP    [::]:5355             *: *
UDP    [::1]:1900           *: *
UDP    [::1]:54240          *: *
UDP    [fe80::9517:fc79:abaa:aeb2%10]:1900 *: *
UDP    [fe80::9517:fc79:abaa:aeb2%10]:2177 *: *
UDP    [fe80::9517:fc79:abaa:aeb2%10]:54239 *: *

```

Command: netstat -r

Description: Displays the system routing table.

Screenshot:

```

C:\Users\User>netstat -r
=====
Interface List
15...84 ba 59 8b fc 58 .....Intel(R) Ethernet Connection (23) I219-V
8...72 08 94 c6 78 7b .....Microsoft Wi-Fi Direct Virtual Adapter
10...70 08 94 c6 78 7b .....Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.1.1      192.168.1.8      50
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.1.0                255.255.255.0    On-link          192.168.1.8      306
192.168.1.8                255.255.255.255  On-link          192.168.1.8      306
192.168.1.255              255.255.255.255  On-link          192.168.1.8      306
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          192.168.1.8      306
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          192.168.1.8      306
=====
Persistent Routes:
None

IPv6 Route Table

```

DNS & Name Resolution

Command: nslookup www.google.com

Description: Queries DNS servers to resolve a domain name to an IP address.

Screenshot:

```
C:\Users\User>nslookup www.google.com
Server: UnKnown
Address: 192.168.1.1

Non-authoritative answer:
Name: www.google.com
Addresses: 2a00:1450:4019:812::2004
          142.250.202.4
```

ARP & Routing

Command: arp -a

Description: Displays the ARP cache mapping IP addresses to MAC addresses.

Screenshot:

```
C:\Users\User>arp -a

Interface: 192.168.1.8 --- 0xa
Internet Address      Physical Address      Type
192.168.1.1           a8-41-22-de-51-18    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

Command: route print

Description: Displays current routing table entries.

Screenshot:

```
C:\Users\User>route print
=====
Interface List
15...84 ba 59 8b fc 58 .....Intel(R) Ethernet Connection (23) I219-V
8...72 08 94 c6 78 7b .....Microsoft Wi-Fi Direct Virtual Adapter
10...70 08 94 c6 78 7b .....Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.1.1      192.168.1.8      50
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.1.0                255.255.255.0    On-link          192.168.1.8      306
192.168.1.8                255.255.255.255  On-link          192.168.1.8      306
192.168.1.255              255.255.255.255  On-link          192.168.1.8      306
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          192.168.1.8      306
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          192.168.1.8      306
=====
Persistent Routes:
None
```

Command: route add 192.168.2.0 mask 255.255.255.0 192.168.1.1

Description: Adds a static route.

Screenshot:

```
C:\Users\User>route add 192.168.2.0 mask 255.255.255.0 192.168.1.1
The requested operation requires elevation.
```

Network Interface & Configuration (netsh)

Command: netsh interface ip show config

Description: Displays IP configuration for all network interfaces.

Screenshot:

```

C:\Users\User>netsh interface ip show config

Configuration for interface "Ethernet"
    DHCP enabled:                Yes
    InterfaceMetric:             5
    DNS servers configured through DHCP: None
    Register with which suffix:   Primary only
    WINS servers configured through DHCP: None

Configuration for interface "Local Area Connection* 1"
    DHCP enabled:                Yes
    InterfaceMetric:             25
    DNS servers configured through DHCP: None
    Register with which suffix:   Primary only
    WINS servers configured through DHCP: None

Configuration for interface "Wi-Fi"
    DHCP enabled:                Yes
    IP Address:                  192.168.1.8
    Subnet Prefix:                192.168.1.0/24 (mask 255.255.255.0)
    Default Gateway:              192.168.1.1
    Gateway Metric:               0
    InterfaceMetric:             50
    DNS servers configured through DHCP: 192.168.1.1
    Register with which suffix:   Primary only
    WINS servers configured through DHCP: 192.168.1.1
    WINS servers configured through DHCP: 192.168.1.1

```

Command: netsh interface ipv4 show dns

Description: Displays DNS configuration.

Screenshot:

```

C:\Users\User>netsh interface ipv4 show dns

Configuration for interface "Ethernet"
    DNS servers configured through DHCP: None
    Register with which suffix:           Primary only

Configuration for interface "Local Area Connection* 1"
    DNS servers configured through DHCP: None
    Register with which suffix:           Primary only

Configuration for interface "Local Area Connection* 2"
    DNS servers configured through DHCP: None
    Register with which suffix:           Primary only

Configuration for interface "Wi-Fi"
    DNS servers configured through DHCP: 192.168.1.1
    Register with which suffix:           Primary only

Configuration for interface "Loopback Pseudo-Interface 1"
    Statically Configured DNS Servers:    None
    Register with which suffix:           Primary only

```

Command: netsh interface ip set address name="Ethernet" static 192.168.1.100 255.255.255.0 192.168.1.1

Description: Sets a static IP address for an interface.

Note: Configuration-changing commands were not executed to avoid modifying system network settings.

Command: netsh interface ip set dns name="Ethernet" static 8.8.8.8

Description: Sets a static DNS server.

Note: Configuration-changing commands were not executed to avoid modifying system network settings.

Wireless Networking

Command: netsh wlan show interfaces

Description: Displays wireless network adapter information.

Screenshot:

```
C:\Users\User>netsh wlan show interfaces

There is 1 interface on the system:

Name                           : Wi-Fi
Description                     : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
GUID                           : 46d048ac-c3d6-460b-8220-95abbf03a42f
Physical address                : 70:08:94:c6:78:7b
Interface type                  : Primary
State                           : connected
SSID                            : oper-wala
AP BSSID                       : a8:41:22:de:51:18
Band                            : 2.4 GHz
Channel                         : 1
Connected Akm-cipher            : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type                    : Infrastructure
Radio type                      : 802.11n
Authentication                  : WPA2-Personal
Cipher                          : CCMP
Connection mode                 : Auto Connect
Receive rate (Mbps)             : 144.4
Transmit rate (Mbps)            : 144.4
Signal                          : 85%
Rssi                            : -54
Profile                         : oper-wala
QoS MSCS Configured             : 0
QoS Map Configured              : 0
QoS Map Allowed by Policy       : 0
```

Command: netsh wlan show profiles

Description: Displays saved Wi-Fi profiles.

Screenshot:

```

C:\Users\User>netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
-----
    <None>

User profiles
-----
    All User Profile      : Galaxy A15
    All User Profile      : Don't touch
    All User Profile      : AndroidShare_62
    All User Profile      : narzo 50
    All User Profile      : TECNO CAMON 40
    All User Profile      : Galaxy s21
    All User Profile      : Galaxy A318336
    All User Profile      : Galaxy S21 FE 5G 97fb
    All User Profile      : DESKTOP-E10FGK9 8790
    All User Profile      : oper-wala
    All User Profile      : Mehar Dawood.
    All User Profile      : Mian's Pixel 9
    All User Profile      : HR
    All User Profile      : Mutahhar Ahmad Bhutta
    All User Profile      : ...
    All User Profile      : Pixel
    All User Profile      : Iphone
    All User Profile      : Oper wala
    All User Profile      : Eu0sa-U0hBUkVpdA==

```

Command: netsh wlan connect name="NetworkName"

Description: Connects to Wi-Fi profile.

Screenshot:

```

C:\Users\User>netsh wlan connect name="Mehar Dawood."
Connection request was completed successfully.

```

Command: netsh wlan disconnect

Description: Disconnects Wi-Fi connection.

Screenshot:

```

C:\Users\User>netsh wlan disconnect
Disconnection request was completed successfully for interface "Wi-Fi".

```

Firewall Management

Command: netsh advfirewall show allprofiles

Description: Displays firewall status.

Screenshot:

```
C:\Users\User>netsh advfirewall show allprofiles

Domain Profile Settings:
-----
State                ON
Firewall Policy      BlockInbound,AllowOutbound
LocalFirewallRules   N/A (GPO-store only)
LocalConSecRules     N/A (GPO-store only)
InboundUserNotification Enable
RemoteManagement    Disable
UnicastResponseToMulticast Enable

Logging:
LogAllowedConnections Disable
LogDroppedConnections Disable
FileName             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize          4096

Private Profile Settings:
-----
State                ON
Firewall Policy      BlockInbound,AllowOutbound
LocalFirewallRules   N/A (GPO-store only)
LocalConSecRules     N/A (GPO-store only)
InboundUserNotification Enable
RemoteManagement    Disable
UnicastResponseToMulticast Enable
```

Command: netsh advfirewall set allprofiles state off

Description: Turns firewall OFF.

Screenshot:

```
C:\Users\User>netsh advfirewall set allprofiles state off
The requested operation requires elevation (Run as administrator).
```

Command: netsh advfirewall set allprofiles state on

Description: Turns firewall ON.

Screenshot:

```
C:\Users\User>netsh advfirewall set allprofiles state on
The requested operation requires elevation (Run as administrator).
```

File Sharing & Network Resources

Command: net use Z: \\servername\sharename

Description: Maps network drive using hostname.

Screenshot:

```
C:\Users\User>net use Z: \\test\share
System error 67 has occurred.

The network name cannot be found.
```

Command: net use Z: \\192.168.1.10\sharename

Description: Maps network drive using IP address.

Screenshot:

```
C:\Users\User>net use Z: \\192.168.1.10\sharename
System error 67 has occurred.

The network name cannot be found.
```

Command: net share sharename=C:\folderpath

Description: Shares local folder.

Screenshot:

```
C:\Users\User>net share sharename=C:\folderpath
System error 5 has occurred.

Access is denied.
```

Command: net view \\servername

Description: Shows shared resources.

Screenshot:

```
C:\Users\User>net view \\servername
System error 53 has occurred.

The network path was not found.
```

Network Services & Statistics

Command: net start servicename

Description: Starts Windows service.

Screenshot:

```
C:\Users\User>net start servicename
The service name is invalid.

More help is available by typing NET HELPMSG 2185.
```

Command: net stop servicename

Description: Stops Windows service.

Screenshot:

```
C:\Users\User>net stop servicename
The service name is invalid.

More help is available by typing NET HELPMSG 2185.
```

Command: net statistics workstation

Description: Displays workstation statistics.

Screenshot:

```
C:\Users\User>net statistics workstation
Workstation Statistics for \\MIAN-ABDULLAH

Statistics since 1/28/2026 11:34:05 AM

Bytes received                               53140
Server Message Blocks (SMBs) received        20
Bytes transmitted                           49782
Server Message Blocks (SMBs) transmitted      0
Read operations                             0
Write operations                             0
Raw reads denied                             0
Raw writes denied                             0

Network errors                               0
Connections made                             0
Reconnections made                             0
Server disconnects                             0

Sessions started                             0
Hung sessions                                0
Failed sessions                               0
Failed operations                             0
Use count                                    11
Failed use count                              0

The command completed successfully.
```

Command: net statistics server

Description: Displays server statistics.

Screenshot:

```
C:\Users\User>net statistics server
The syntax of this command is:

NET STATISTICS
[WORKSTATION]
```

Command: net config server

Description: Shows server configuration.

Screenshot:

```
C:\Users\User>net config server
System error 5 has occurred.

Access is denied.
```

Command: nbtstat -a hostname

Description: Displays NetBIOS statistics.

Screenshot:

```
C:\Users\User>nbtstat -a hostname

Ethernet:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

Wi-Fi:
Node IpAddress: [192.168.1.6] Scope Id: []

    Host not found.

Local Area Connection* 1:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

Local Area Connection* 2:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.
```

PowerShell Network Commands

Configuration & Status

Command: Get-NetAdapter

Description: Lists network adapters.

Screenshot:

```
PS C:\Users\User> Get-NetAdapter
```

Name	InterfaceDescription	ifIndex	Status	MacAddress	LinkSpeed
Ethernet	Intel(R) Ethernet Connection (23) I2...	15	Disconnected	84-BA-59-8B-FC-58	0 bps
Wi-Fi	Realtek RTL8852BE WiFi 6 802.11ax PC...	10	Up	70-08-94-C6-78-7B	144.4 Mbps

Command: Get-NetIPAddress

Description: Shows IP address configuration.

Screenshot:

```
PS C:\Users\User> Get-NetIPAddress
```

IPAddress : fe80::1ba8:a48e:ba6f:af4f%12
InterfaceIndex : 12
InterfaceAlias : Local Area Connection* 2
AddressFamily : IPv6
Type : Unicast
PrefixLength : 64
PrefixOrigin : WellKnown
SuffixOrigin : Link
AddressState : Deprecated
ValidLifetime :
PreferredLifetime :
SkipAsSource : False
PolicyStore : ActiveStore

IPAddress : fe80::ef43:7e1e:eb40:9da2%15
InterfaceIndex : 15
InterfaceAlias : Ethernet
AddressFamily : IPv6
Type : Unicast
PrefixLength : 64
PrefixOrigin : WellKnown
SuffixOrigin : Link
AddressState : Deprecated
ValidLifetime :
PreferredLifetime :
SkipAsSource : False

Command: Get-NetIPConfiguration

Description: Displays detailed interface configuration.

Screenshot:

```
PS C:\Users\User> Get-NetIPConfiguration

InterfaceAlias      : Wi-Fi
InterfaceIndex      : 10
InterfaceDescription : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
NetProfile.Name      : oper-wala
IPv4Address          : 192.168.1.6
IPv6DefaultGateway   : fe80::1
IPv4DefaultGateway   : 192.168.1.1
DNSServer            : 192.168.1.1

InterfaceAlias      : Ethernet
InterfaceIndex      : 15
InterfaceDescription : Intel(R) Ethernet Connection (23) I219-V
NetAdapter.Status    : Disconnected
```

Command: Get-NetRoute

Description: Displays routing table in PowerShell.

Screenshot:

```
PS C:\Users\User> Get-NetRoute
```

ifIndex	DestinationPrefix	NextHop	RouteMetric	ifMetric	PolicyStore
12	255.255.255.255/32	0.0.0.0	256	25	Ac
8	255.255.255.255/32	0.0.0.0	256	25	Ac
10	255.255.255.255/32	0.0.0.0	256	50	Ac
15	255.255.255.255/32	0.0.0.0	256	5	Ac
1	255.255.255.255/32	0.0.0.0	256	75	Ac
12	224.0.0.0/4	0.0.0.0	256	25	Ac
8	224.0.0.0/4	0.0.0.0	256	25	Ac
10	224.0.0.0/4	0.0.0.0	256	50	Ac
15	224.0.0.0/4	0.0.0.0	256	5	Ac
1	224.0.0.0/4	0.0.0.0	256	75	Ac
10	192.168.1.255/32	0.0.0.0	256	50	Ac
10	192.168.1.6/32	0.0.0.0	256	50	Ac
10	192.168.1.0/24	0.0.0.0	256	50	Ac
1	127.255.255.255/32	0.0.0.0	256	75	Ac
1	127.0.0.1/32	0.0.0.0	256	75	Ac
1	127.0.0.0/8	0.0.0.0	256	75	Ac
10	0.0.0.0/0	192.168.1.1		50	Ac
12	ff00::/8	::	256	25	Ac
8	ff00::/8	::	256	25	Ac

Connectivity & DNS

Command: Test-Connection www.google.com

Description: PowerShell connectivity test.

Screenshot:

```
PS C:\Users\User> Test-Connection www.google.com
```

Source	Destination	IPv4Address	IPv6Address	Bytes	Time(ms)
MIAN-ABDULLAH	www.google.com	142.250.187.68		32	33
MIAN-ABDULLAH	www.google.com	142.250.187.68		32	34
MIAN-ABDULLAH	www.google.com	142.250.187.68		32	122
MIAN-ABDULLAH	www.google.com	142.250.187.68		32	33

Command: Resolve-DnsName www.google.com

Description: Resolves DNS name using PowerShell.

Screenshot:

```
PS C:\Users\User> Resolve-DnsName www.google.com
```

Name	Type	TTL	Section	IPAddress
www.google.com	AAAA	2143	Answer	2001:4860:482a:7700::
www.google.com	AAAA	2143	Answer	2001:4860:482c:7700::
www.google.com	AAAA	2143	Answer	2001:4860:4829:7700::
www.google.com	AAAA	2143	Answer	2001:4860:4828:7700::
www.google.com	AAAA	2143	Answer	2001:4860:4826:7700::
www.google.com	AAAA	2143	Answer	2001:4860:482d:7700::
www.google.com	AAAA	2143	Answer	2001:4860:482b:7700::
www.google.com	AAAA	2143	Answer	2001:4860:4827:7700::
www.google.com	A	563	Answer	142.250.187.68

Command: Get-DnsClientCache

Description: Displays DNS client cache.

Screenshot:

```
PS C:\Users\User> Get-DnsClientCache
```

Entry	RecordName	Record Type	Status	Section	TimeTo Live	Data Length	Data
fe3cr.delivery.mp.micr...	fe3cr.delivery.mp.micr...	CNAME	Success	Answer	916	8	fe3.delivery.mp.microsoft...
fe3cr.delivery.mp.micr...	fe3.delivery.mp.micros...	CNAME	Success	Answer	916	8	glb.cws.prod.dcat.dsp.tra...
fe3cr.delivery.mp.micr...	glb.cws.prod.dcat.dsp...	A	Success	Answer	916	4	40.69.42.241
fe3cr.delivery.mp.micr...	fe3cr.delivery.mp.micr...	CNAME	Success	Answer	916	8	fe3.delivery.mp.microsoft...
fe3cr.delivery.mp.micr...	fe3.delivery.mp.micros...	CNAME	Success	Answer	916	8	glb.cws.prod.dcat.dsp.tra...
fe3cr.delivery.mp.micr...	glb.cws.prod.dcat.dsp...	A	Success	Answer	916	4	40.69.42.241
blz04pap002.storage.li...	blz04pcor002-meta.fe.1...	CNAME	Success	Answer	1695	8	blz04pcor002-meta.fe.1drv...
blz04pap002.storage.li...	odc-blz04pap002-meta-g...	CNAME	Success	Answer	1695	8	odc-blz04pap002-meta-geo...
blz04pap002.storage.li...	odc-blz04pap002-meta-b...	CNAME	Success	Answer	1695	8	odc-blz04pap002-meta-brs...
blz04pap002.storage.li...	ln-0001.ln-dc-msedge.n...	CNAME	Success	Answer	1695	8	ln-0001.ln-dc-msedge.net...
blz04pap002.storage.li...	ln-0001.ln-msedge.net	A	Success	Answer	1695	4	150.171.22.11
download.mcafee.com	download.mcafee.com	CNAME	Success	Answer	665	8	download-r53.awsconsumer...
download.mcafee.com	download-r53.awsconsum...	CNAME	Success	Answer	665	8	download.mcafee.com.edges...
download.mcafee.com	download.mcafee.com.ed...	CNAME	Success	Answer	665	8	a2047.x.akamai.net
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.16
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.10
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.18
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.75
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.8
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.11
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.74
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.73
download.mcafee.com	a2047.x.akamai.net	A	Success	Answer	665	4	103.86.38.17

Firewall & Interface Control

Command: Get-NetFirewallProfile

Description: Displays firewall profile configuration.

Screenshot:

```

PS C:\Users\User> Get-NetFirewallProfile

Name           : Domain
Enabled        : True
DefaultInboundAction : NotConfigured
DefaultOutboundAction : NotConfigured
AllowInboundRules  : NotConfigured
AllowLocalFirewallRules : NotConfigured
AllowLocalIPsecRules : NotConfigured
AllowUserApps     : NotConfigured
AllowUserPorts    : NotConfigured
AllowUnicastResponseToMulticast : NotConfigured
NotifyOnListen    : True
EnableStealthModeForIPsec : NotConfigured
LogFileName       : %systemroot%\system32\LogFiles\Firewall\pfirewall.log
LogMaxSizeKilobytes : 4096
LogAllowed        : False
LogBlocked        : False
LogIgnored        : NotConfigured
DisabledInterfaceAliases : {NotConfigured}

Name           : Private
Enabled        : True
DefaultInboundAction : NotConfigured
DefaultOutboundAction : NotConfigured
AllowInboundRules  : NotConfigured
AllowLocalFirewallRules : NotConfigured
AllowLocalIPsecRules : NotConfigured
AllowUserApps     : NotConfigured

```

Command: `Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False`

Description: Disables firewall via PowerShell.

Screenshot:

```

PS C:\Users\User> Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
Set-NetFirewallProfile : Access is denied.
At line:1 char:1
+ Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_NetFirewal...rofile?Public):root/standardci...FirewallProfile)
[Set-NetFirewallProfile], CimException
+ FullyQualifiedErrorId : Windows System Error 5,Set-NetFirewallProfile

Set-NetFirewallProfile : Access is denied.
At line:1 char:1
+ Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_NetFirewal...rofile?Domain):root/standardci...FirewallProfile)
[Set-NetFirewallProfile], CimException
+ FullyQualifiedErrorId : Windows System Error 5,Set-NetFirewallProfile

Set-NetFirewallProfile : Access is denied.
At line:1 char:1
+ Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_NetFirewal...ofile?Private):root/standardci...FirewallProfile)
[Set-NetFirewallProfile], CimException
+ FullyQualifiedErrorId : Windows System Error 5,Set-NetFirewallProfile

```

Command: `Enable-NetAdapter -Name "Ethernet"`

Description: Enables network adapter.

Screenshot:

```

PS C:\Users\User> Enable-NetAdapter -Name "Ethernet"
PS C:\Users\User> |

```

Command: Disable-NetAdapter -Name "Ethernet"

Description: Disables network adapter.

Screenshot:

```
PS C:\Users\User> Disable-NetAdapter -Name "Ethernet"

Confirm
Are you sure you want to perform this action?
Disable-NetAdapter 'Ethernet'
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "Y"): Y
Disable-NetAdapter : Access is denied.
At line:1 char:1
+ Disable-NetAdapter -Name "Ethernet"
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_NetAdapter...Mian-Abdullah):ROOT/StandardCimv2/MSFT_NetAdapter
[Disable-NetAdapter], CimException
+ FullyQualifiedErrorId : Windows System Error 5,Disable-NetAdapter
```

Advanced Configuration

Command: New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.1.100 -PrefixLength 24 -DefaultGateway 192.168.1.1

Description: Assigns new static IP address.

Screenshot:

```
PS C:\Users\User> New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.1.100 -PrefixLength 24 -DefaultGateway 192.168.1.1
New-NetIPAddress : Access is denied.
At line:1 char:1
+ New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.1.100 ...
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_NetIPAddress:ROOT/StandardCimv2/MSFT_NetIPAddress) [New-NetIPAd
ress], CimException
+ FullyQualifiedErrorId : Windows System Error 5,New-NetIPAddress
```

Command: Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 8.8.8.8

Description: Sets manual DNS server.

Screenshot:

```
PS C:\Users\User> Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 8.8.8.8
Set-DnsClientServerAddress : Access to a CIM resource was not available to the client.
At line:1 char:1
+ Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAdresse ...
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_DNSClientS...temName = "23"):ROOT/StandardCi...ntServerAddress)
[Set-DnsClientServerAddress], CimException
+ FullyQualifiedErrorId : MI_RESULT 2,Set-DnsClientServerAddress

Set-DnsClientServerAddress : Access to a CIM resource was not available to the client.
At line:1 char:1
+ Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAdresse ...
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (MSFT_DNSClientS...stemName = "2"):ROOT/StandardCi...ntServerAddress)
[Set-DnsClientServerAddress], CimException
+ FullyQualifiedErrorId : MI_RESULT 2,Set-DnsClientServerAddress
```

Command: Get-NetTCPConnection

Description: Displays active TCP connections.

Screenshot:


```
PS C:\Users\User> Get-NetTCPConnection
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting
-----	-----	-----	-----	-----	-----
::1	56842	::	0	Listen	
::	49676	::	0	Listen	
::1	49671	::	0	Listen	
::	49670	::	0	Listen	
::	49669	::	0	Listen	
::	49668	::	0	Listen	
::	49665	::	0	Listen	
::	49664	::	0	Listen	
::1	42050	::	0	Listen	
::	445	::	0	Listen	
::	135	::	0	Listen	
0.0.0.0	64695	0.0.0.0	0	Bound	
0.0.0.0	62929	0.0.0.0	0	Bound	
0.0.0.0	62524	0.0.0.0	0	Bound	
0.0.0.0	60271	0.0.0.0	0	Bound	
0.0.0.0	60115	0.0.0.0	0	Bound	
0.0.0.0	57216	0.0.0.0	0	Bound	
0.0.0.0	57215	0.0.0.0	0	Bound	
0.0.0.0	57214	0.0.0.0	0	Bound	
0.0.0.0	57213	0.0.0.0	0	Bound	
0.0.0.0	57212	0.0.0.0	0	Bound	
0.0.0.0	57211	0.0.0.0	0	Bound	
0.0.0.0	57210	0.0.0.0	0	Bound	
0.0.0.0	57209	0.0.0.0	0	Bound	
0.0.0.0	55755	0.0.0.0	0	Bound	

Help & Documentation

Command: `command /?`

Description: Displays command help.

Screenshot:

```
C:\Users\User>ipconfig /?

USAGE:
    ipconfig [/allcompartments] [/? | /all |
        /renew [adapter] | /release [adapter] |
        /renew6 [adapter] | /release6 [adapter] |
        /flushdns | /displaydns | /registerdns |
        /showclassid adapter |
        /setclassid adapter [classid] |
        /showclassid6 adapter |
        /setclassid6 adapter [classid] ]

where
    adapter          Connection name
                     (wildcard characters * and ? allowed, see examples)

Options:
    /?              Display this help message
    /all            Display full configuration information.
    /release        Release the IPv4 address for the specified adapter.
    /release6       Release the IPv6 address for the specified adapter.
    /renew          Renew the IPv4 address for the specified adapter.
    /renew6         Renew the IPv6 address for the specified adapter.
    /flushdns       Purges the DNS Resolver cache.
    /registerdns     Refreshes all DHCP leases and re-registers DNS names
    /displaydns     Display the contents of the DNS Resolver Cache.
    /showclassid    Displays all the dhcp class IDs allowed for adapter.
    /setclassid     Modifies the dhcp class id.
    /showclassid6   Displays all the IPv6 DHCP class IDs allowed for adapter.
```

Command: `Get-Help Get-NetIPAddress`

Description: Displays PowerShell help documentation.

Screenshot:


```
PS C:\Users\User> Get-Help Get-NetIPAddress
```

NAME

Get-NetIPAddress

SYNTAX

```
Get-NetIPAddress [[-IPAddress] <string[]>] [-InterfaceIndex <uint32[]>] [-InterfaceAlias <string[]>]
[-AddressFamily {IPv4 | IPv6}] [-Type {Unicast | Anycast}] [-PrefixLength <byte[]>] [-PrefixOrigin {Other | Manual
| WellKnown | Dhcp | RouterAdvertisement}] [-SuffixOrigin {Other | Manual | WellKnown | Dhcp | Link | Random}]
[-AddressState {Invalid | Tentative | Duplicate | Deprecated | Preferred}] [-ValidLifetime <timespan[]>]
[-PreferredLifetime <timespan[]>] [-SkipAsSource <bool[]>] [-AssociatedIPInterface
<CimInstance#MSFT_NetIPInterface>] [-PolicyStore <string>] [-IncludeAllCompartments] [-CimSession <CimSession[]>]
[-ThrottleLimit <int>] [-AsJob] [<CommonParameters>]
```

ALIASES

None

REMARKS

Get-Help cannot find the Help files for this cmdlet on this computer. It is displaying only partial help.
-- To download and install Help files for the module that includes this cmdlet, use Update-Help.